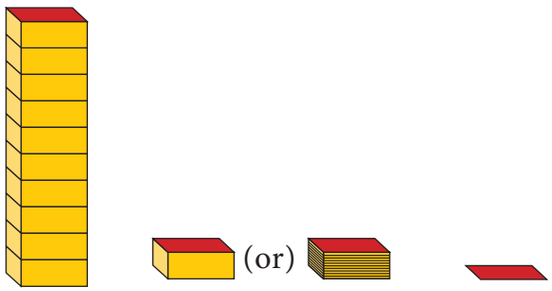
1.2 Representing a Decimal Number

(i) Observe the following pictures which represent tens, ones and tenths



For example, the decimal number 3.2 is 3 ones and 2 tenths that can be represented using the pictorial representations as follows.



Similarly any decimal number can be represented as above

Try this

Represent the following decimal numbers pictorially.

(i) 5 ones and 3 tenths

(ii) 6 tenths

(iii) 7 ones and 9 tenths

(iv) 6 ones and 4 tenths

(v) Seven tenths

We have studied about place value of digits of a number in primary classes. Now let us extend our learning of place values to decimal digits of a number. Let us recall the expanded form of a number. Consider the number 3768.

The expanded form of 3768 is given by $3\times1000+7\times100+6\times10+8$.

Now, let us consider a decimal number 235.68

Expressing the decimal number in the expanded form, we get

$$235.68 = 200+30+5+\frac{6}{10}+\frac{8}{100}$$
$$= 2\times100+3\times10+5\times1+6\times\frac{1}{10}+8\times\frac{1}{100}$$

Think

The above expression can also be expressed in terms of powers of 10 as follows $235.68 = 2\times10^{2}+3\times10^{1}+5\times10^{0}+6\times10^{-1}+8\times10^{-2}$. Hence, in any number, when we move towards right from one digit to the next, the place value of a digit is divided by 10^{1}.

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